AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 16, 2016

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In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (30) Consider the motion of a spacecraft subject to the gravity of the Earth modelled as a point mass and its own thrust. Let m be the mass of the spacecraft, $\mu = GM$ be the gravitational parameter of the Earth where G is Newton's gravitational constant and M is the mass of the Earth, $\vec{r}$ be the position vector of the spacecraft, and $\vec{F}$ be the thrust.

   A. (5) Let $\vec{r} = xi + yj + zk$, $\vec{F} = F_x i + F_y j + F_z k$ and the state vector $\vec{x} = [x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6]^T$ $= [x \ \dot{x} \ y \ \dot{y} \ z \ \dot{z}]^T$ be defined in the Cartesian coordinate system. Describe the equations of motion in the Cartesian coordinate system in state space form.

   B. (10) Let $\vec{r} = r\vec{e}_r$ and $\vec{F} = F_r\vec{e}_r + F_v\vec{e}_v + F_\lambda\vec{e}_\lambda$ be defined in the spherical coordinate system with their components in the order of radial, longitudinal, and latitude direction, and $\vec{\omega}$ be the angular velocity vector of the spacecraft. Describe the equations of motion in the spherical coordinate system.

   C. (15) Linearize the radial (first) equations of (B) with respect to the following operation point:

$$\begin{cases} r \equiv R > 0 \\ v = \sqrt{\dfrac{\mu}{R^3}}t + C, \ C = \text{constant} \\ \lambda \equiv 0 \end{cases}$$

$r = R + \Delta r$

$v = \sqrt{\frac{\mu}{R^3}}\, t + \Delta v$

$\lambda = 0 + \Delta\lambda$

2. (20) Consider the Euler's rotational equations of motion represented in the principal axes of body-fixed frame:

$$\begin{cases} I_x\dot{\omega}_x = (I_y - I_z)\omega_y\omega_z + M_x \\ I_y\dot{\omega}_y = (I_z - I_x)\omega_x\omega_z + M_y \\ I_z\dot{\omega}_z = (I_x - I_y)\omega_x\omega_y + M_z \end{cases}$$

   A. (5) For a general asymmetric spacecraft, show that $(\omega_x \equiv 0, \omega_y \equiv 0, \omega_z \equiv \bar{\omega}_z = \text{constant})$ is an equilibrium solution with no external moments.

   B. (5) Linearize the Euler's rotational equations of motion with respect to the equilibrium solution given in (A).

   C. (10) Derive the characteristic equation for two non-trivial equations obtained in (B) and their associated eigenvalues case-by-case (i.e., depending on the magnitude of moment of inertia component).

(5)

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3. (25) An inertially symmetric rigid spacecraft has a diagonal inertia matrix given by $I = diag(200, 200, 200) kg - m^2$. It is desired that the spacecraft perform a rotational rest-to-rest maneuver that transfers the spacecraft from its initial attitude given by a rotational matrix representation $R = \exp[S(a)]$, where the unit vector $a = (0.424, -0.566, 0.707)$, to the final attitude given by the rotational matrix R that is the identity. The rotational maneuver should be a rotation about the fixed axis defined by the unit vector a (it is an Euler rotation), and the rotational acceleration about that fixed axis should be either $+0.01\, rad/sec^2$ or $-0.01\, rad/sec^2$. Assume gravity gradient moments can be ignored.

   A. Give an analytical expression for the time dependence of each component of the spacecraft angular velocity vector expressed in the body-fixed coordinate frame.

   B. Give an analytical expression for the time dependence of each component of the control moment vector expressed in the body-fixed coordinate frame.

   C. How long is a time period required to complete the rotational maneuver?

4. (25) Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y$$

   where the orbital velocity vector $\omega_0 = 0.0012\, rad/sec$, the spacecraft principal moments of inertia are $I = diag(150, 200, 100) kg - m^2$. In this equation $\Delta\theta$ denotes the pitch angle and M_y denotes the pitch axis control moment. It is desired to perform a rest-to-rest maneuver using the pitch axis torques that transfers the initial pitch angle $\Delta\theta = 45$degrees to the final pitch angle $\Delta\theta = 0$degrees.

   It is claimed that this rest-to-rest maneuver can be performed using a single impulse supplied by the pitch axis control moment.

   Is this claim true or false?

   If the claim is true, then describe at what time instant the impulse should be applied and describe the required change in the pitch rate due to this impulse.

   If the claim is false, justify that it is not possible to accomplish the maneuver using a single impulse.